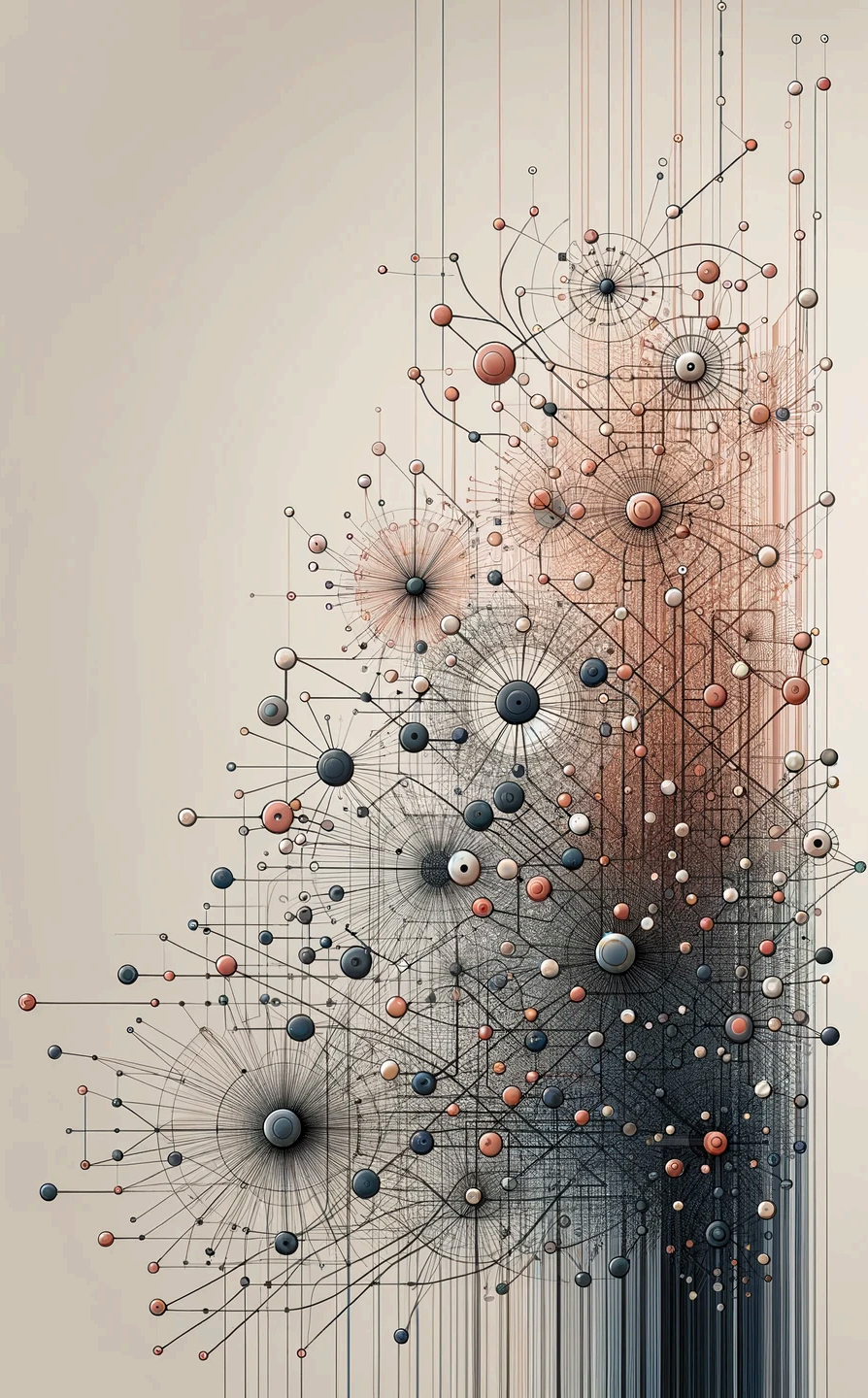


Analisi e Visualizzazione delle Reti Complesse

NS23 - Computational Epidemiology - Network Models

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Beyond Homogeneous Mixing

Limitations of Compartmental Models

- Assume **homogeneous mixing**: "Anyone can infect anyone with equal probability."
- Ignore critical heterogeneity in:
 - Contact patterns
 - Interaction types
 - Social structures
 - Geographic distribution
- Consequences:
 - **Overestimates** epidemic size in clustered populations
 - **Underestimates** early spread in heterogeneous networks
 - **Inaccurately predicts** effects of targeted interventions
 - Cannot explain **superspreading** phenomena

Network Representation in Epidemiology

Why networks matter

- **Natural solution** to the limitations of homogeneous mixing
- Explicitly model **who can infect whom**
- Capture **heterogeneity** in contact patterns
- Enable **realistic modeling** of transmission dynamics

Formal representation

- Network $G = (V, E)$ where:
 - V = set of nodes (individuals)
 - E = set of edges (contacts)
- Each node i has a discrete state: S , I , or R
- Infection spreads along network edges

From Compartmental to Network Models

Building Network Structure into Epidemic Models

- Key question: How can we extend epidemic models to networks while balancing **complexity** and **accuracy**?

Three Approaches:

- **Homogeneous mean-field** : Simplest extension that incorporates average connectivity
 - Preserves mathematical tractability while adding basic network properties
- **Heterogeneous mean-field** : Accounts for variation in connectivity
 - Critical for realistic modeling of hub effects in scale-free networks
- **Individual-based probabilistic models** : Uses full network structure
 - Captures stochastic effects and enables targeted interventions

Approach 1: Homogeneous Mean-Field Approximation

Main idea: Extend the classic SIR model to networks by using the network's average degree.

Assumption: All individuals are statistically equivalent and have, on average, the same number of contacts.

- Simple and mathematically tractable
- Ignores differences in individual connectivity

SIR Model Equations (Homogeneous Mean-Field)

$$\begin{aligned}\frac{dS(t)}{dt} &= -\beta \langle k \rangle \frac{S(t)I(t)}{N} \\ \frac{dI(t)}{dt} &= \beta \langle k \rangle \frac{S(t)I(t)}{N} - \gamma I(t) \\ \frac{dR(t)}{dt} &= \gamma I(t)\end{aligned}$$

Where:

- $S(t), I(t), R(t)$: numbers of susceptible, infected, and recovered individuals at time t
- N : total population size
- $\langle k \rangle$: average degree (average number of contacts per person)
- β : transmission rate per contact
- γ : recovery rate

Digging Deeper (Homogeneous Mean-Field)

1. **Contact structure:** Each person has $\langle k \rangle$ contacts on average.
2. **Probability of infectious contact:** The chance a contact is infected is $\frac{I(t)}{N}$.
3. **Expected infectious contacts:** Each susceptible has $\langle k \rangle \frac{I(t)}{N}$ infectious contacts.
4. **Force of infection:** Each infectious contact transmits at rate β .
5. **Total new infections:** $\beta \langle k \rangle \frac{S(t)I(t)}{N}$.

Key Quantities (Homogeneous Mean-Field)

- Basic reproduction number: $R_0 = \frac{\beta \langle k \rangle}{\gamma}$
- Critical transmission rate: $\beta_c = \frac{\gamma}{\langle k \rangle}$
- Early-stage growth exponent: $\alpha = \beta \langle k \rangle - \gamma = \gamma(R_0 - 1)$

Interpretation & Consequences (Homogeneous Mean-Field)

- R_0 : Increases linearly with average degree $\langle k \rangle$. More contacts per person mean faster and larger outbreaks.
- β_c : Higher average degree lowers the epidemic threshold; diseases can spread more easily in denser networks.
- **Growth exponent α** : Epidemics grow exponentially if $R_0 > 1$; the speed of growth depends on both transmission and network density.
- **Limitation**: This approach cannot capture the effects of hubs or degree variability—only the average matters.

Approach 2: Heterogeneous Mean-Field

Main idea: Account for differences in individual connectivity by grouping nodes by degree.

Assumption: Individuals with the same degree behave identically, but nodes can have different degrees.

- Captures the effect of hubs and degree variability
- More realistic for heterogeneous networks

SIR Model Equations (Heterogeneous Mean-Field)

$$\begin{aligned}\frac{dS_k(t)}{dt} &= -\beta k S_k(t) \Theta(t) \\ \frac{dI_k(t)}{dt} &= \beta k S_k(t) \Theta(t) - \gamma I_k(t) \\ \frac{dR_k(t)}{dt} &= \gamma I_k(t)\end{aligned}$$

Where:

- $S_k(t)$, $I_k(t)$, $R_k(t)$: numbers of susceptible, infected, and recovered nodes of degree k at time t
- k : degree (number of contacts)
- $\Theta(t)$: probability a random contact points to an infected node

Calculating $\Theta(t)$

$$\Theta(t) = \frac{\sum_{k'} k' P(k') I_{k'}(t)}{\langle k \rangle N}$$

- k' : possible degrees in the network
- $P(k')$: fraction of nodes with degree k'
- $I_{k'}(t)$: number of infected nodes of degree k' at time t
- $\langle k \rangle$: average degree
- N : total population

Note:

- $\Theta(t)$ is the same for all k (uncorrelated networks).
- An uncorrelated network is a network where the degree of a node's neighbor is independent of the node's own degree.

Why does $\Theta(t)$ have this form?

- **Goal:** We want the probability that, if we follow a random edge in the network, we arrive at an infected node.
- **Why not just $I(t)/N$?** In heterogeneous networks, nodes with more connections are more likely to be reached by following a random edge.
- **How it works:**
 - $k'P(k')$ = probability that a random edge leads to a node of degree k' (since high-degree nodes have more edges).
 - $I_{k'}(t)$ = fraction of degree k' nodes that are infected.
 - Multiply and sum over all degrees: $\sum_{k'} k'P(k')I_{k'}(t)$ gives the total "weight" of infected ends of edges.
 - Divide by total number of edges ($\langle k \rangle N$) to get a probability.
- **Intuition:** The formula gives more weight to infected nodes with many connections, reflecting their higher chance of being encountered along a random edge.

Digging Deeper (Heterogeneous Mean-Field)

1. **Degree classes:** Nodes are grouped by degree k .
2. **Contact structure:** A node with degree k has k contacts.
3. **Probability of infectious contact:** Each contact is with an infected node with probability $\Theta(t)$.
4. **Expected infectious contacts:** $k\Theta(t)$ for a degree- k node.
5. **Force of infection:** $\beta k\Theta(t)$ for each susceptible of degree k .

Key Quantities (Heterogeneous Mean-Field)

- Basic reproduction number: $R_0 = \frac{\beta}{\gamma} \frac{\langle k^2 \rangle}{\langle k \rangle}$
- Critical transmission rate: $\beta_c = \frac{\gamma \langle k \rangle}{\langle k^2 \rangle}$
- Early-stage growth exponent: $\alpha = \beta \frac{\langle k^2 \rangle}{\langle k \rangle} - \gamma = \gamma(R_0 - 1)$

Understanding the $\frac{\langle k^2 \rangle}{\langle k \rangle}$ Factor

- $\langle k^2 \rangle$: Average of the square of the degree (captures variance in connectivity)
- $\langle k \rangle$: Average degree
- **Interpretation:** High variance (large $\langle k^2 \rangle$) means more hubs, which increases R_0 and lowers the epidemic threshold.

Interpretation & Consequences (Heterogeneous Mean-Field)

- R_0 : Strongly affected by degree variance ($\langle k^2 \rangle$). Networks with hubs (high $\langle k^2 \rangle$) have much higher R_0 .
- β_c : Can become extremely small in networks with large degree variance (e.g., scale-free networks), meaning even weakly transmissible diseases can cause epidemics.
- **Growth exponent α** : Early epidemic growth is much faster in heterogeneous networks due to the influence of highly connected nodes.
- **Consequence**: Targeting hubs for intervention is much more effective than random strategies.

Approach 3: Individual-Based SIR Models on Networks

Main idea: Track the state of each individual node using the exact network structure.

Assumption: Each node can have a unique set of neighbors; the full adjacency matrix is used.

- Most detailed and realistic
- Captures stochastic effects and local network structure

SIR Model Equations (Individual-Based)

$$\frac{dS_i(t)}{dt} = -\beta S_i(t) \sum_j A_{ij} I_j(t)$$

$$\frac{dI_i(t)}{dt} = \beta S_i(t) \sum_j A_{ij} I_j(t) - \gamma I_i(t)$$

$$\frac{dR_i(t)}{dt} = \gamma I_i(t)$$

Where:

- $S_i(t), I_i(t), R_i(t)$: probability node i is susceptible, infected, or recovered at time t
- A_{ij} : adjacency matrix (1 if i and j are connected, 0 otherwise)

Digging Deeper (Individual-Based)

1. **Node-level tracking:** Each node i has its own equations.
2. **Contact structure:** The sum $\sum_j A_{ij} I_j(t)$ counts infected neighbors of node i .
3. **Force of infection:** $\beta \sum_j A_{ij} I_j(t)$ for node i .
4. **Stochasticity:** Can be simulated as a random process or as probabilities.

Key Quantities (Individual-Based)

- **Basic reproduction number:** $R_0 = \frac{\beta}{\gamma} \lambda_{max}$
 - λ_{max} : largest eigenvalue of the adjacency matrix A
 - The largest eigenvalue measures how strongly connected and influential the network is as a whole.
- **Per-node basic reproduction number:** $R_0(i) = \frac{\beta}{\gamma} v_i$
 - Eigenvector centrality (v_i) measures how well-connected node i is, not just by counting its direct neighbors, but also by considering how central those neighbors are.
- **Critical transmission rate:** $\beta_c = \frac{\gamma}{\lambda_{max}}$
- **Early-stage growth exponent:** $\alpha = \beta \lambda_{max} - \gamma = \gamma(R_0 - 1)$

Interpretation & Consequences (Individual-Based)

- R_0 : Determined by the most "central" or influential nodes and the overall network structure (not just degree distribution).
- β_c : Sensitive to specific patterns of connectivity; a single well-connected node can lower the threshold for the whole network.
- **Growth exponent α** : Epidemics grow fastest in regions of the network associated with the largest eigenvector components (often hubs or bridges).
- **Consequence**: This approach allows precise identification of critical nodes for intervention and captures effects of clustering, community structure, and real-world network features.

Comparing the Three Approaches

Model	Network Structure	R_0	Critical Threshold β_c
Homogeneous MF	Average degree	$\frac{\beta \langle k \rangle}{\gamma}$	$\frac{\gamma}{\langle k \rangle}$
Heterogeneous MF	Degree distribution	$\frac{\beta}{\gamma} \frac{\langle k^2 \rangle}{\langle k \rangle}$	$\frac{\gamma \langle k \rangle}{\langle k^2 \rangle}$
Individual-based	Full adjacency matrix	$\frac{\beta}{\gamma} \lambda_{max}$	$\frac{\gamma}{\lambda_{max}}$

Summary: As we move from homogeneous to individual-based models, we gain realism and accuracy at the cost of complexity and data requirements.

Applications and Interventions

Network models allow testing of targeted intervention strategies:

- **Random vaccination:** Choose nodes randomly
- **Degree-based:** Target high-degree nodes (hubs)
- **Betweenness-based:** Target nodes connecting communities
- **Acquaintance:** Vaccinate randomly selected neighbors of random nodes

Network topology strongly influences the optimal strategy choice!

Q&A

